Senior *DeFi Smart Contract* Engineer

Location: Remote / Poland (International Team)
Employment Type: Full-time / Part-time / Contract
Department: *Blockchain* Engineering & *DeFi* Infrastructure
Reports To: Head of Engineering / *Blockchain* Architect
Seniority: Senior

About the Company

We are preparing to establish an independent software development company in Poland, focused on building secure, scalable financial technology and *blockchain*-based applications.

Our planned products and technical initiatives include:

- ***DeFi* & Decentralized Exchanges:** Decentralized *trading* applications, liquidity management, staking, yield-generating protocols, and *smart contract* integrations.
- **Digital Asset Infrastructure:** *Token* management, *cryptocurrency* transactions, *wallet*s, and *blockchain* settlement services.
- **Fintech & Payment Solutions:** Payment processing, stable*coin* integrations, digital asset transfers, and financial transaction infrastructure.
- **Real Estate Investment Platforms:** Investment marketplaces and *token*ized real-world asset applications.
- **Cross-Chain Services:** *Blockchain* interoperability, asset transfers, and integrations with multiple networks.

Our goal is to combine reliable software engineering with *blockchain* technology to create practical financial applications.

We are building an international engineering team and looking for experienced *blockchain* engineers who enjoy solving complex technical challenges and developing secure financial infrastructure.

About the Role

We are looking for a **Senior *DeFi Smart Contract* Engineer** to design, develop, test, and maintain *smart contract*s for decentralized financial applications.

This is a hands-on engineering role focused on *Solidity* development, *DeFi* protocol integrations, *smart contract* security, and reliable on-chain financial operations.

You will collaborate with *blockchain* architects, backend engineers, frontend developers, security specialists, and product teams to develop *smart contract* systems supporting digital asset transactions, decentralized exchanges, staking, liquidity management, and other financial workflows.

The ideal candidate has substantial experience building production *smart contract*s, understands the mechanics of decentralized finance, and can make sound engineering decisions involving security, gas efficiency, upgradeability, and protocol interoperability.

We value engineers who understand not only how to write *Solidity* code, but also how *DeFi* protocols behave under real market conditions.

What You'll Do

Smart Contract Development

- Design, implement, and maintain *smart contract*s using *Solidity*.
- Develop *smart contract* components supporting decentralized financial applications.
- Implement *token* management, staking, rewards distribution, and asset transfer functionality.
- Design secure contract interfaces for interaction with backend services and *Web3* applications.
- Apply established *Ethereum* standards and development patterns where appropriate.
- Optimize *smart contract*s for correctness, security, maintainability, and gas efficiency.
- Develop reusable *Solidity* libraries and modular contract components.
- Participate in architecture discussions and *smart contract* design reviews.

DeFi Protocol Engineering

- Develop and integrate *smart contract* components for decentralized exchanges, liquidity pools, and staking applications.
- Work with *DeFi* primitives involving AMMs, *swap*s, liquidity provision, vaults, and yield distribution.
- Design and implement reliable asset accounting and fee calculation mechanisms.
- Integrate established *DeFi* protocols through audited interfaces and SDKs.
- Develop contract interactions supporting portfolio management and digital asset operations.
- Evaluate protocol dependencies, economic assumptions, and potential integration risks.
- Collaborate with backend engineers to connect on-chain execution with off-chain systems.

- Support transaction settlement and lifecycle management for *blockchain* financial applications.

Smart Contract Security

- Apply secure coding practices to prevent common *smart contract* vulnerabilities.
- Identify and mitigate reentrancy, access-control, arithmetic, signature, and authorization risks.
- Review contract logic for accounting errors, precision loss, and unsafe state transitions.
- Evaluate oracle dependencies, price manipulation risks, and flash-loan attack surfaces.
- Implement appropriate permission management, emergency controls, and protocol safeguards.
- Participate in internal security reviews and external *smart contract* audits.
- Address audit findings and validate security improvements.
- Develop comprehensive test suites covering normal operations, edge cases, and adversarial scenarios.
- Apply defense-in-depth principles to contracts managing financial assets.

Testing, Verification & Deployment

- Build automated *smart contract* tests using Foundry, Hardhat, or comparable frameworks.
- Develop unit, integration, fuzz, and invariant tests for critical protocol functionality.
- Use static analysis and security tools to identify potential vulnerabilities.
- Create reproducible deployment scripts and contract verification workflows.
- Support testnet deployments and controlled production releases.
- Monitor deployed contracts and investigate unexpected on-chain behavior.
- Improve CI/CD pipelines for *smart contract* testing and deployment.
- Maintain clear documentation covering contract behavior, dependencies, and operational procedures.

Blockchain Integrations

- Integrate *smart contract*s with *EVM*-compatible *blockchain* networks.
- Collaborate with backend teams on *blockchain* event in*dex*ing, transaction submission, and confirmation tracking.
- Develop interfaces for *wallet* integrations and decentralized applications.
- Support contract interactions involving ERC-20 *token*s and other relevant *token* standards.
- Evaluate *blockchain* network characteristics, transaction costs, and execution limitations.
- Contribute to the design of on-chain and off-chain workflows.
- Work with infrastructure engineers to improve reliability of *blockchain* RPC and in*dex*ing integrations.
- Support cross-chain integrations when required by product needs.

Technical Ownership & Collaboration

- Take ownership of *smart contract* components from initial design to production deployment.
- Review *Solidity* code and provide constructive technical feedback.
- Mentor junior and mid-level *blockchain* engineers.
- Collaborate with product teams to evaluate technical feasibility and implementation trade-offs.
- Participate in protocol architecture and security discussions.
- Document important technical decisions and contract assumptions.
- Help establish engineering standards for *Solidity* development, testing, and deployment.
- Stay informed about relevant *DeFi* protocol developments and emerging security risks.

Required Skills & Experience

- 5+ years of professional software engineering experience, including substantial hands-on *blockchain* development.
- 3+ years of professional *Solidity* and *EVM smart contract* development experience.
- Proven experience designing, implementing, and deploying production *smart contract*s.
- S*trong* understanding of *Ethereum*, *EVM* execution, transactions, events, storage, and gas mechanics.
- Experience working with *DeFi* protocols such as decentralized exchanges, staking systems, lending applications, or liquidity management platforms.
- S*trong* understanding of *Solidity* security vulnerabilities and secure *smart contract* design.
- Experience with Foundry, Hardhat, or equivalent *smart contract* development frameworks.
- Practical experience developing automated tests for *smart contract*s.
- Familiarity with fuzz testing, invariant testing, and static analysis.
- Knowledge of ERC-20, ERC-721, ERC-4626, and other relevant *Ethereum* standards.
- Experience integrating *smart contract*s with external protocols and *Web3* applications.
- Understanding of financial asset accounting, transaction flows, and on-chain state management.
- Familiarity with *blockchain* RPC interfaces, event logs, in*dex*ing, and transaction monitoring.
- S*trong* understanding of Git, code reviews, CI/CD, and modern software development practices.
- Ability to independently investigate complex *smart contract* bugs and security issues.
- S*trong* communication skills and experience collaborating with distributed engineering teams.

Nice to Have (Differentiators)

- Experience developing AMMs, liquidity pools, vaults, or lending protocol components.
- Familiarity with *Uniswap*, Curve, *Aave*, Compound, Balancer, or comparable *DeFi* protocols.
- Experience with ERC-4626 *token*ized vaults and advanced *DeFi* accounting.
- Knowledge of flash loans, MEV, oracle manipulation, liquidation mechanics, and economic attacks.
- Experience with Chainlink or comparable oracle infrastructure.
- Understanding of EIP-712, signature-based authorization, permits, and meta-transactions.
- Familiarity with account abstraction and ERC-4337.
- Experience building *smart contract*s for stable*coin* payments and digital asset settlement.
- Understanding of cross-chain bridges, messaging protocols, and interoperability risks.
- Experience with Layer 2 networks such as Arbitrum, Optimism, Base, or zk*EVM*-based environments.
- Familiarity with upgradeable contract patterns and proxy architectures.
- Experience with *smart contract* formal verification or advanced property-based testing.
- Participation in *smart contract* audits, bug bounty programs, or competitive security reviews.
- Open-source contributions to *Solidity* libraries, *DeFi* protocols, or developer tooling.
- Familiarity with *Rust*, Go, TypeScript, or Python for *blockchain* integrations.
- Experience with *token*ized real-world assets or institutional *DeFi* systems.

Education

- Bachelor's or Master's degree in Computer Science, Software Engineering, Mathematics, Information Security, or a related technical discipline.
- Equivalent professional experience and demonstrated *smart contract* engineering expertise are equally valued.
- Relevant open-source contributions, production deployments, and technical achievements are highly appreciated.

What Success Looks Like

First 3 Months

- Understand the planned *DeFi* and *blockchain* product architecture.
- Review technical requirements, *smart contract* designs, and major security considerations.
- Contribute to initial contract implementations and protocol integrations.

- Establish comprehensive testing practices for assigned components.
- Collaborate with *blockchain* architects and engineers on technical design decisions.
- Identify security risks and recommend practical improvements.

Within 6 Months

- Independently deliver production-ready *smart contract* components.
- Implement reliable *DeFi* integrations and financial transaction workflows.
- Strengthen automated testing, fuzzing, and security review processes.
- Support contract deployments, monitoring, and incident-response readiness.
- Contribute to architectural improvements and technical documentation.
- Mentor other engineers and participate in code reviews.

Long-Term

- Help develop secure and maintainable *DeFi* infrastructure.
- Improve the reliability and efficiency of *smart contract*s handling digital assets.
- Support the company's expansion into decentralized *trading*, payments, and *token*ized financial applications.
- Establish s*tron*g security and engineering practices across *smart contract* development.
- Contribute to protocol architecture as products and technical requirements evolve.

Why Join Us?

- Participate in building *DeFi* and *blockchain* financial applications from an early stage.
- Work on meaningful engineering challenges involving *smart contract*s, digital assets, and financial transactions.
- Help shape architecture and security practices rather than only implementing isolated features.
- Collaborate with experienced engineers across *blockchain*, backend, frontend, and infrastructure teams.
- Work in an international, remote-friendly environment.
- Explore full-time, part-time, or contract collaboration based on mutual preferences.
- Gain opportunities to contribute to decentralized finance, stable*coin* payments, and real-world asset applications.
- Maintain hands-on technical ownership while contributing to broader engineering decisions.

Ideal Candidate

We are looking for an experienced *Solidity* engineer who combines s*tron*g *smart contract* development skills with a practical understanding of decentralized finance.

You should be comfortable designing complex financial contracts, reviewing protocol architecture, identifying security risks, and building automated testing strategies.

The ideal candidate understands how *smart contract* decisions affect user funds, financial accounting, and overall protocol safety.

We value engineers who prioritize correctness, security, maintainability, and thoughtful technical decision-making over unnecessary complexity.

Previous experience developing production *DeFi* applications, participating in security audits, or contributing to open-source protocols is particularly valuable.

The strongest Senior DeFi Smart Contract Engineer will combine Solidity expertise, DeFi protocol knowledge, security-conscious engineering, and hands-on experience delivering reliable blockchain financial systems.